Data Analytics Internship – Task 3

Sales Performance Analysis

Submitted to: Alfido Tech

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Domain: Data Analytics Internship
Tools Used: Python, Pandas, Matplotlib, Google Colab

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1. Introduction

This report presents an analysis of website-related user activity data. The objective is to understand user behavior, engagement patterns, and interaction trends. Since the dataset does not contain direct website metrics such as bounce rate and session duration, proxy metrics have been used to derive meaningful insights.

The analysis helps in understanding user engagement, popular content, and traffic distribution, which can support decision-making for improving website performance.

2. Dataset Description

The dataset includes the following columns:

- Event
- Date
- Country
- City
- Artist
- Album
- Track
- LinkID

These attributes are used to analyze user interactions, geographic distribution, and content engagement.

3. Data Cleaning

The following preprocessing steps were performed:

- Removed duplicate records
- Converted date column into datetime format
- Checked and handled missing values
- Standardized column formats for analysis

4. Key Metrics

Task3_Website_Traffic_Analysis.ipynb

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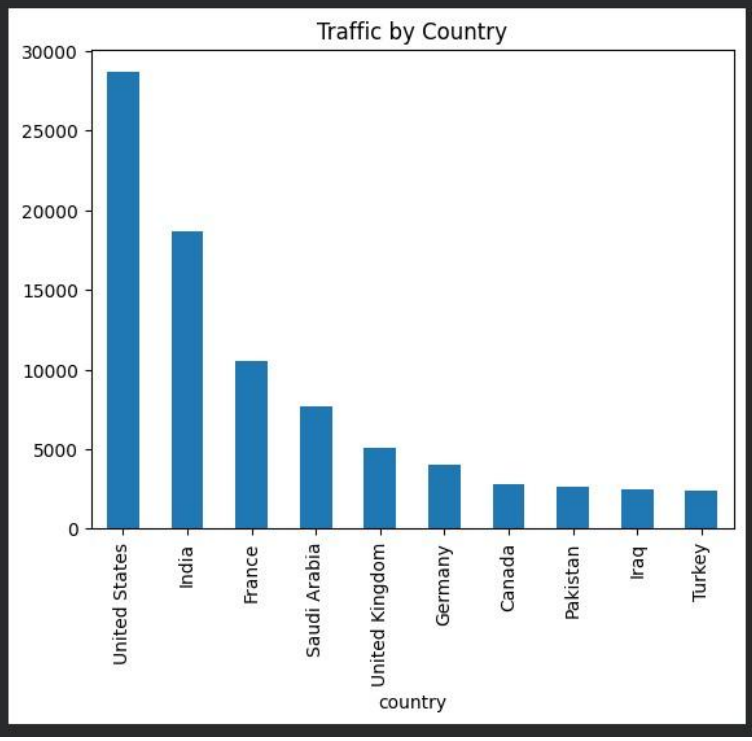
```python
# ==============================
# 📌 1. IMPORT LIBRARIES
# ==============================
import pandas as pd
import matplotlib.pyplot as plt

# ==============================
# 📌 2. UPLOAD FILE
# ==============================
from google.colab import files
uploaded = files.upload()

file_name = list(uploaded.keys())[0]

# ==============================
# 📌 3. LOAD DATA
# ==============================
df = pd.read_csv(file_name, encoding='latin1')

# ==============================
# 📌 4. DATA CLEANING
# ==============================
df.drop_duplicates(inplace=True)

# Check missing values
print("Missing Values:\n", df.isnull().sum())

# ==============================
# 📌 5. BASIC INFO
# ==============================
print("\nDataset Info:\n")
print(df.info())
```

```
Choose Files traffic.csv
traffic.csv(text/csv) - 30630222 bytes, last modified: 3/18/2026 - 100% done
Saving traffic.csv to traffic.csv
Missing Values:
 event         0
date           0
country        5
city           5
artist        28
album          4
track          4
isrc        6306
linkid         0
dtype: int64

Dataset Info:

<class 'pandas.core.frame.DataFrame'>
Index: 122567 entries, 0 to 226274
Data columns (total 9 columns):
 #   Column   Non-Null Count   Dtype
---  ------   --------------   -----
 0   event    122567 non-null  object
 1   date     122567 non-null  object
 2   country  122562 non-null  object
 3   city     122562 non-null  object
 4   artist   122539 non-null  object
 5   album    122563 non-null  object
 6   track    122563 non-null  object
 7   isrc     116261 non-null  object
 8   linkid   122567 non-null  object
dtypes: object(9)
memory usage: 9.4+ MB
None
```

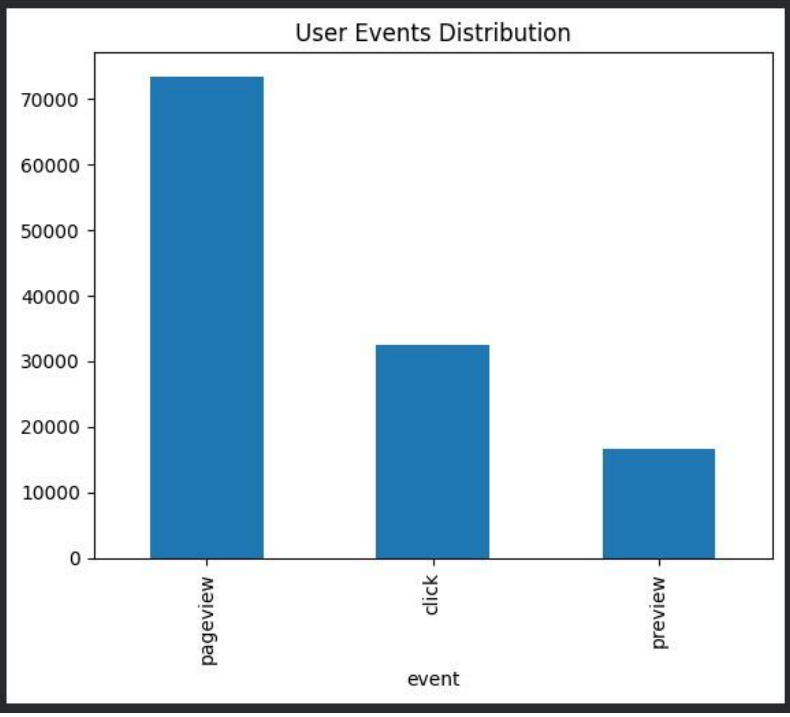
```
[16] ✓ 0s
print("Total Sessions:", len(df))

Total Sessions: 122567

[17] ✓ 0s
print("Total Users (approx):", df['city'].nunique())

Total Users (approx): 11993

[18] ✓ 0s
print("Event Distribution:\n", df['event'].value_counts(normalize=True)*100)

Event Distribution:
 event
pageview    59.852978
click       26.515294
preview     13.631728
Name: proportion, dtype: float64

[19] ✓ 0s
print("Average Daily Activity:", df.groupby('date').size().mean())

Average Daily Activity: 17509.571428571428
```

5. Data Visualization

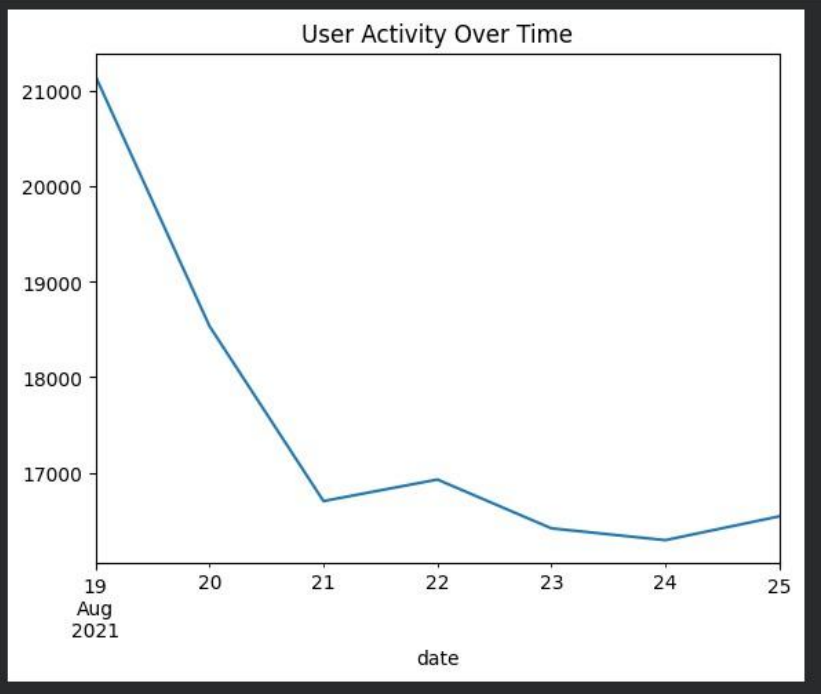
```
df['country'].value_counts().head(10).plot(kind='bar', title='Traffic by Country')
plt.show()
```



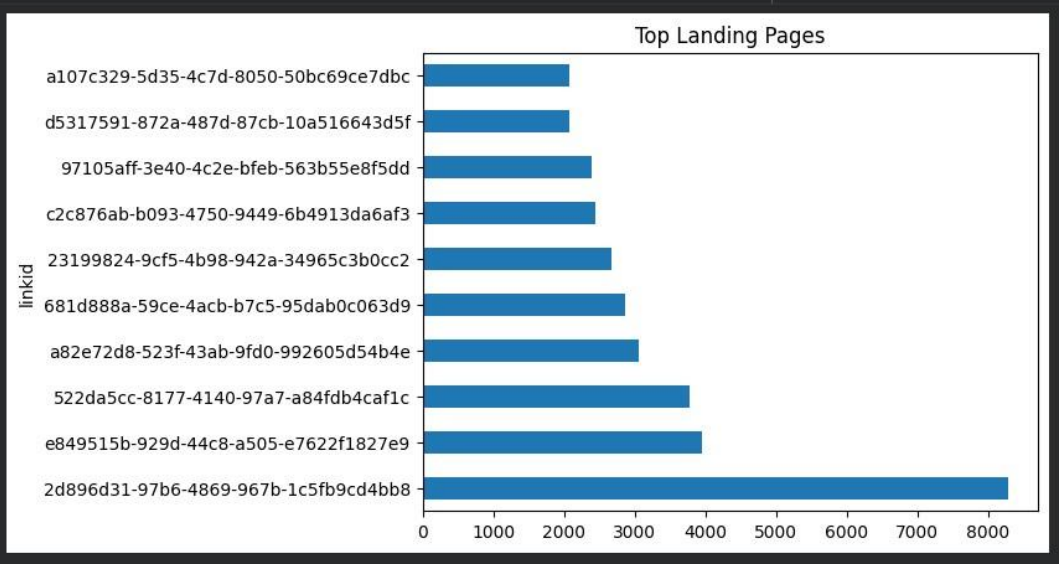

```
df['event'].value_counts().plot(kind='bar', title='User Events Distribution')
plt.show()
```

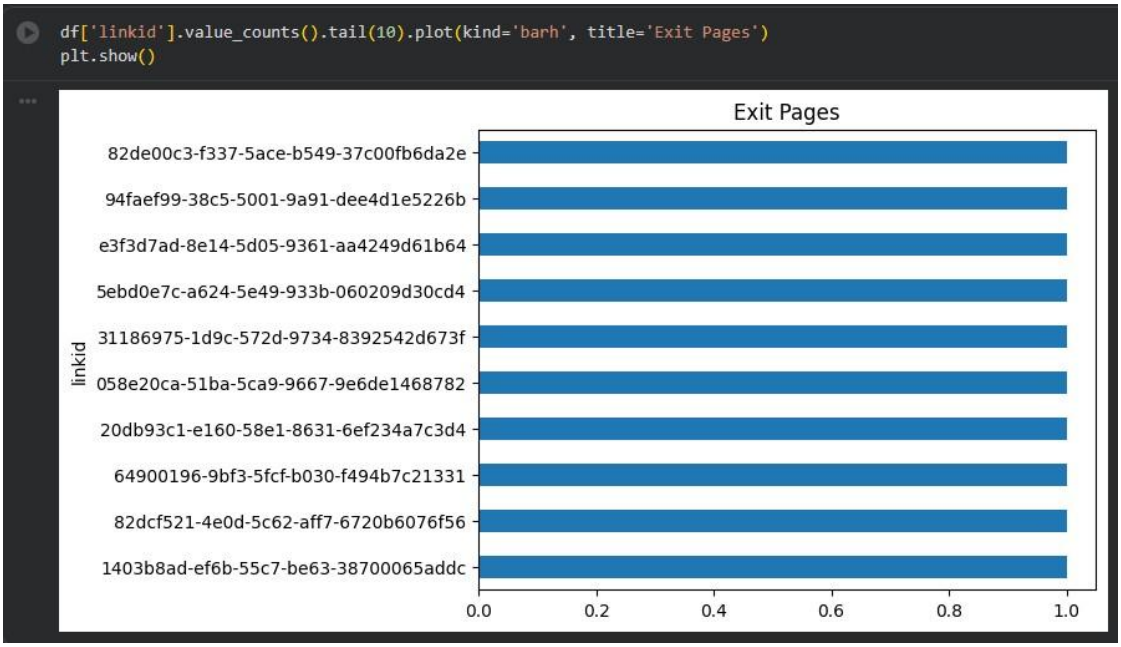


```
df['date'] = pd.to_datetime(df['date'], errors='coerce')

df.groupby('date').size().plot(title='User Activity Over Time')
plt.show()
```




```
df['linkid'].value_counts().head(10).plot(kind='barh', title='Top Landing Pages')
plt.show()
```





6. Key Insights

Majority of traffic comes from specific countries

User interaction is concentrated around certain events

User interaction is concentrated around certain events

User activity varies over time indicating engagement trends

Some links receive higher traffic showing popular content

7. Recommendations

1. Focus marketing efforts on high-traffic countries
2. Optimize popular content to improve engagement
3. Improve low-performing links to reduce drop-offs
4. Monitor user activity trends for better planning
5. Personalize content based on user interaction behaviour

8. Project Links

https://github.com/Swayamchavan/InternSpark-Internship